

1. Problem Statement

Users often need to check multiple shopping platforms to find the best price for the same product. This process is time-consuming, and differences in discounts, quantities, delivery charges, and availability make comparison difficult. The proposed application provides a single platform to compare product prices across services such as Blinkit, Instamart, BigBasket, and JioMart, helping users quickly identify the most cost-effective option.

2. Functional Requirements

- **Product Search:** Users can search for products by name or category.
- **Price Comparison:** Display prices of the same product across different platforms.
- **Product Matching:** Identify and match equivalent products across platforms.
- **Sorting & Filtering:** Sort by price, discount, availability, or platform.
- **Product Details:** Show price, quantity, discount, delivery charges, and availability.
- **Best Deal:** Highlight the platform offering the best overall price.
- **Price History:** Store and display previous prices for comparison.
- **User Management:** Optional login/signup for saving products and preferences.

3. Scope

- Compare products across multiple platforms such as Blinkit, Instamart, BigBasket, and JioMart.
- Provide a centralized interface for searching and comparing products.
- Compare **price, quantity, discounts, delivery charges, and availability**.
- Maintain price history to identify price changes and trends.
- Support location-based comparison where platform pricing varies by location.
- Future scope can include **AI-based product matching, personalized recommendations, price-drop alerts, and mobile/PWA support**.

4. Constraints

- Product prices and availability may change frequently and may vary by location.
- Access to real-time data may depend on the availability of official APIs or permitted data sources.
- Different platforms may use different product names, quantities, brands, and packaging, making product matching challenging.
- Delivery charges, discounts, and offers may vary based on user location and account.
- The accuracy of comparisons depends on the freshness and reliability of collected data.
- Platform policies and technical restrictions may limit automated data collection.